

# *Deep Learning - Theory and Practice*

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Basics of Machine Learning

23-01-2020

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<http://leap.ee.iisc.ac.in/sriram/teaching/DL20/>

[deeplearning.cce2020@gmail.com](mailto:deeplearning.cce2020@gmail.com)



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# Problems in Decision Theory

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- ❖ Decision Theory
  - ❖ Inference problem
    - ❖ Finding the joint density  $p(\mathbf{x}, \mathbf{t})$
  - ❖ Decision problem
    - ❖ Using the inference to make the classification or regression decision

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# Decision Problem - Classification

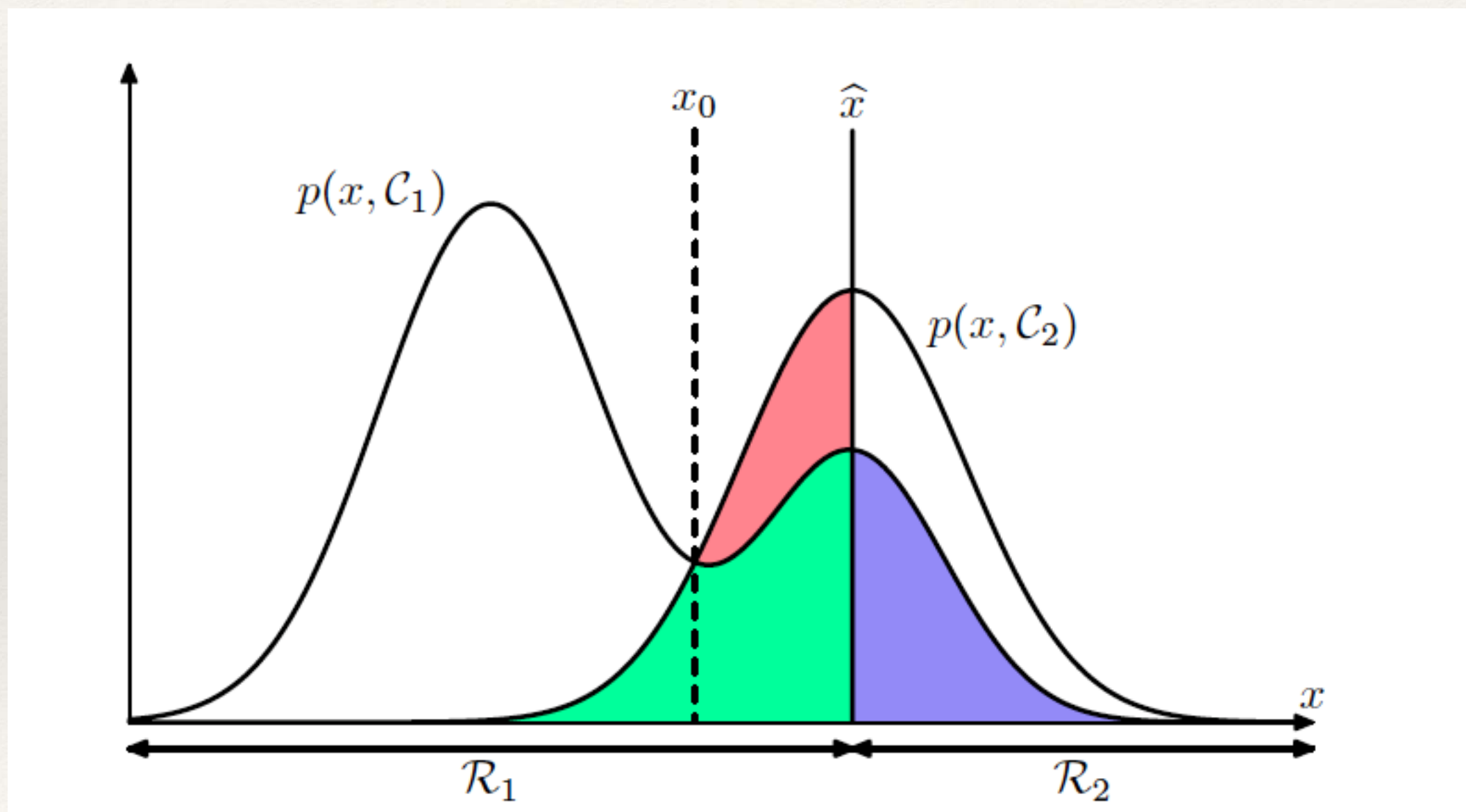
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- ❖ Minimizing the mis-classification error
- ❖ Decision based on maximum posteriors

$$\mathit{argmax}_j p(C_j|\mathbf{x})$$

- ❖ Loss matrix
  - ❖ Can be used for non uniform error weighting.

# Decision Theory



# Approaches for Inference and Decision

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I. Finding the joint density from the data.

$$p(C_k|\mathbf{x}) \propto p(\mathbf{x}|C_k)p(C_k)$$

II. Finding the posteriors directly.

**Neural Networks**

III. Using discriminant functions for classification.

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# Advantages of Posteriors

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- ❖ Minimizing the risk
- ❖ Reject Option
- ❖ Combining models
- ❖ Compensating for class priors

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# Loss Function for Regression

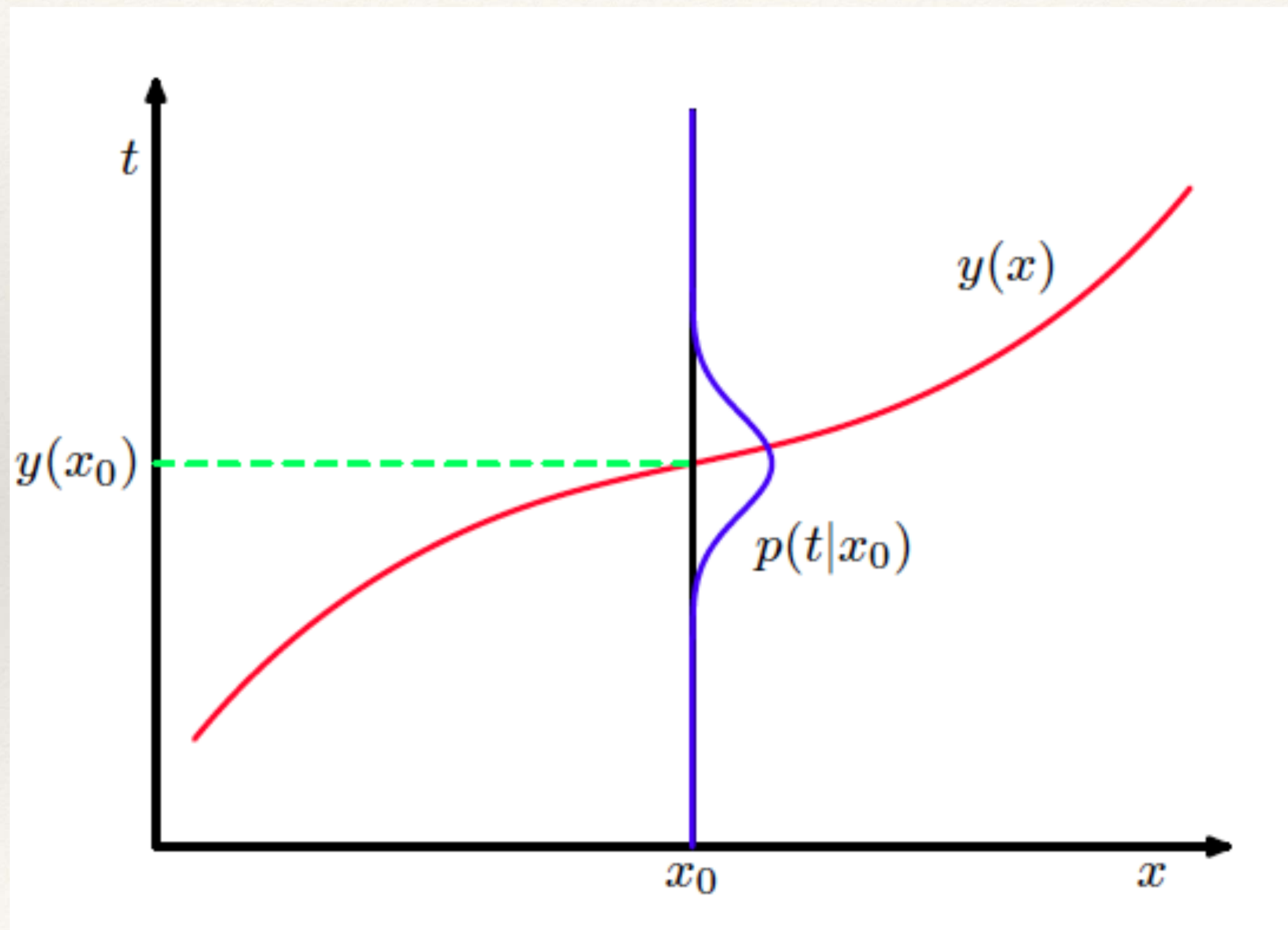
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- ❖ With a mean square error loss

$$\mathbb{E}[L] = \iint \{y(\mathbf{x}) - t\}^2 p(\mathbf{x}, t) \, d\mathbf{x} \, dt.$$

- ❖ The problem boils down to conditional expectation of the data given the

# Regression Problem



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# Matrix Derivatives

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$$\left(\frac{\partial \mathbf{a}}{\partial x}\right)_i = \frac{\partial a_i}{\partial x}$$

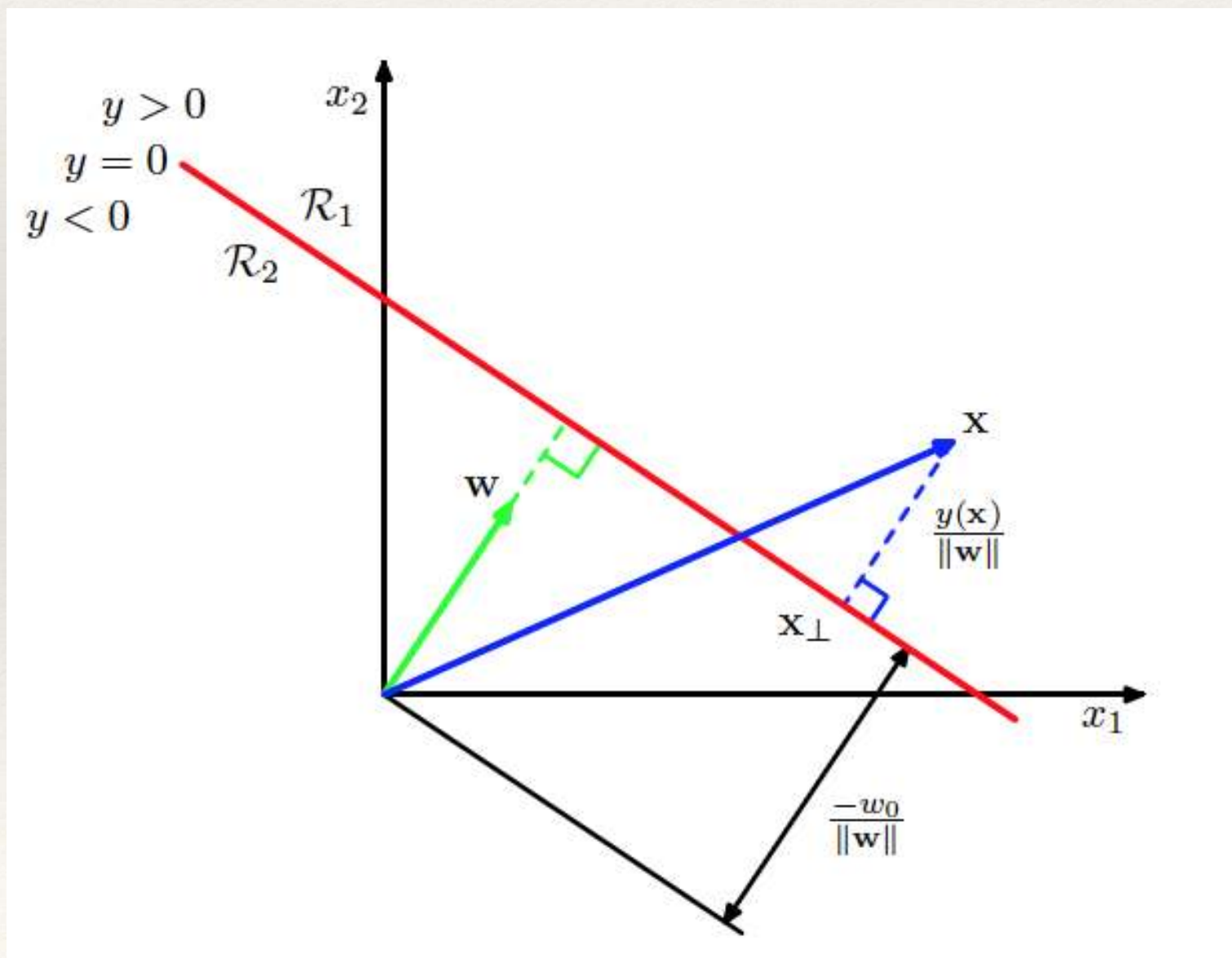
$$\left(\frac{\partial x}{\partial \mathbf{a}}\right)_i = \frac{\partial x}{\partial a_i}$$

$$\left(\frac{\partial \mathbf{a}}{\partial \mathbf{b}}\right)_{ij} = \frac{\partial a_i}{\partial b_j}.$$

# Linear Models for Classification

- ❖ Optimize a modified cost function

$$y(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0$$



# Least Squares for Classification

- ❖ K-class classification problem

$$y_k(\mathbf{x}) = \mathbf{w}_k^T \mathbf{x} + w_{k0}$$

$$\mathbf{y}(\mathbf{x}) = \widetilde{\mathbf{W}}^T \widetilde{\mathbf{x}}$$

- ❖ With 1-of-K hot encoding, and least squares regression

$$E_D(\widetilde{\mathbf{W}}) = \frac{1}{2} \text{Tr} \left\{ (\widetilde{\mathbf{X}} \widetilde{\mathbf{W}} - \mathbf{T})^T (\widetilde{\mathbf{X}} \widetilde{\mathbf{W}} - \mathbf{T}) \right\}$$

# Principal Component Analysis

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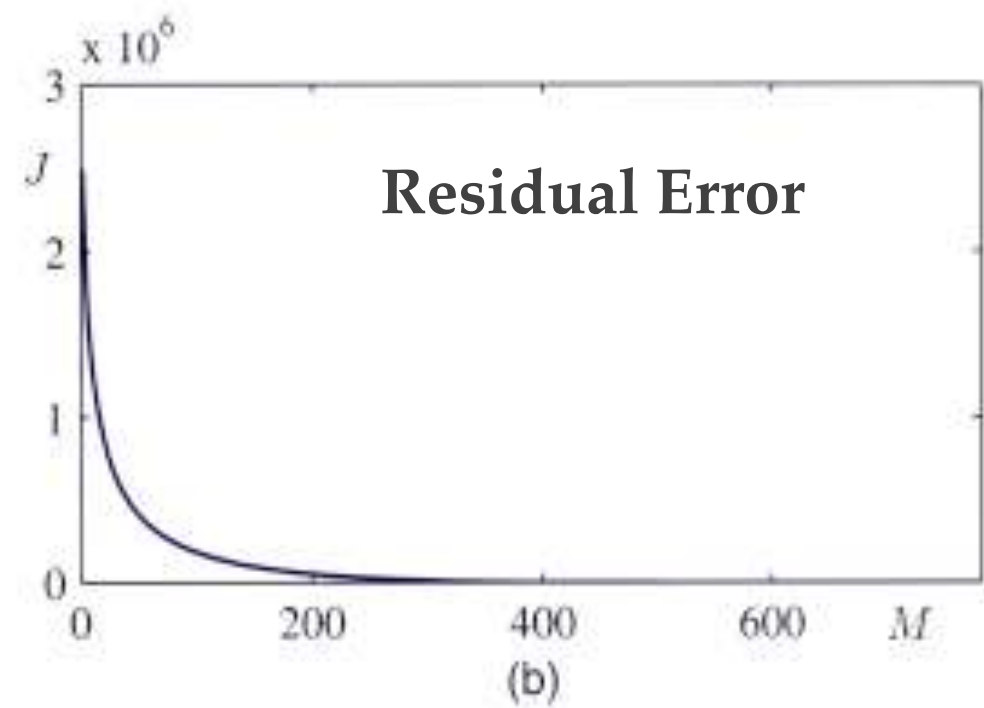
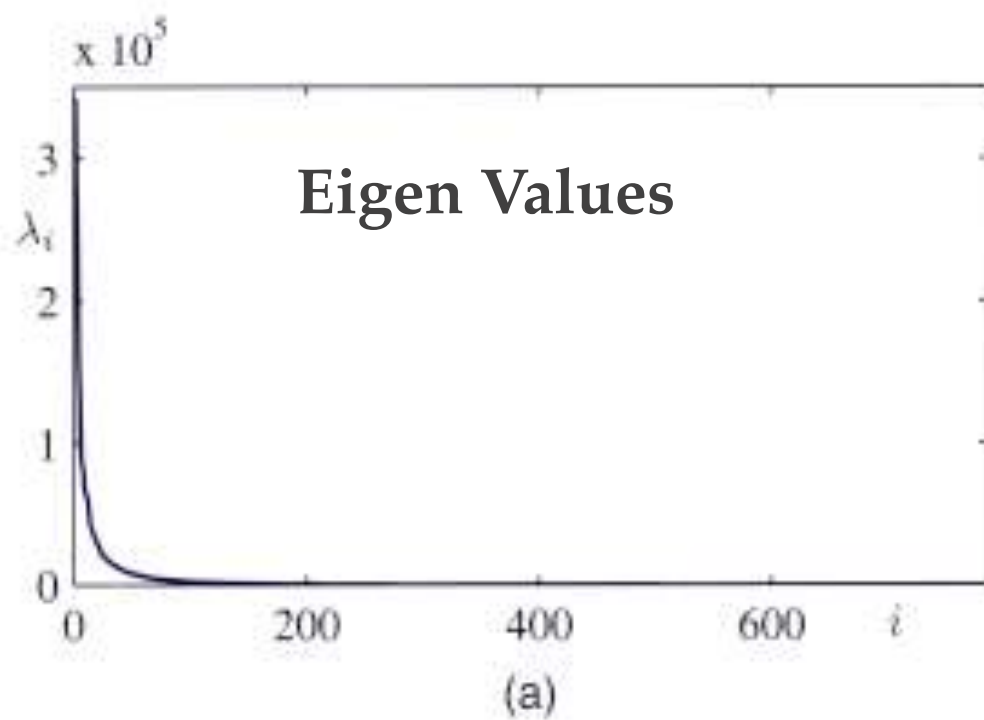
- ❖ First  $M$  eigenvectors of data covariance matrix

$$S = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T$$

- ❖ Residual error from PCA

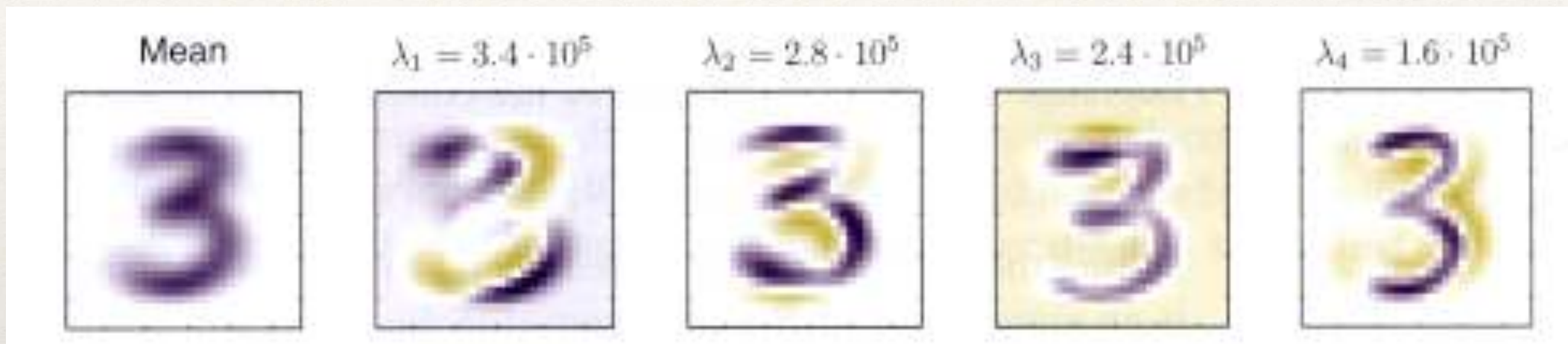
$$J = \sum_{i=M+1}^D \lambda_i$$

# PCA

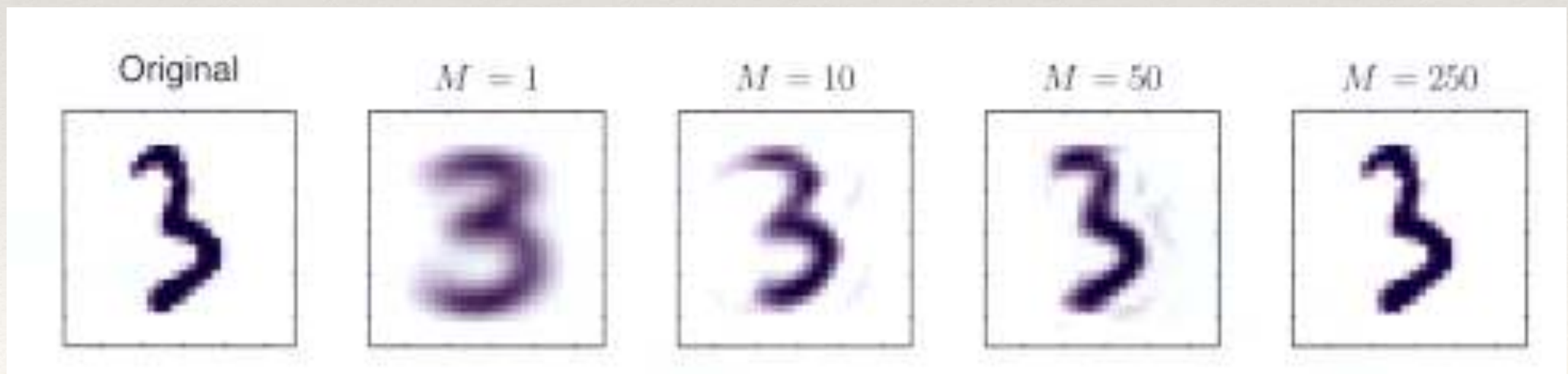


# PCA - Reconstruction

## Eigenvectors

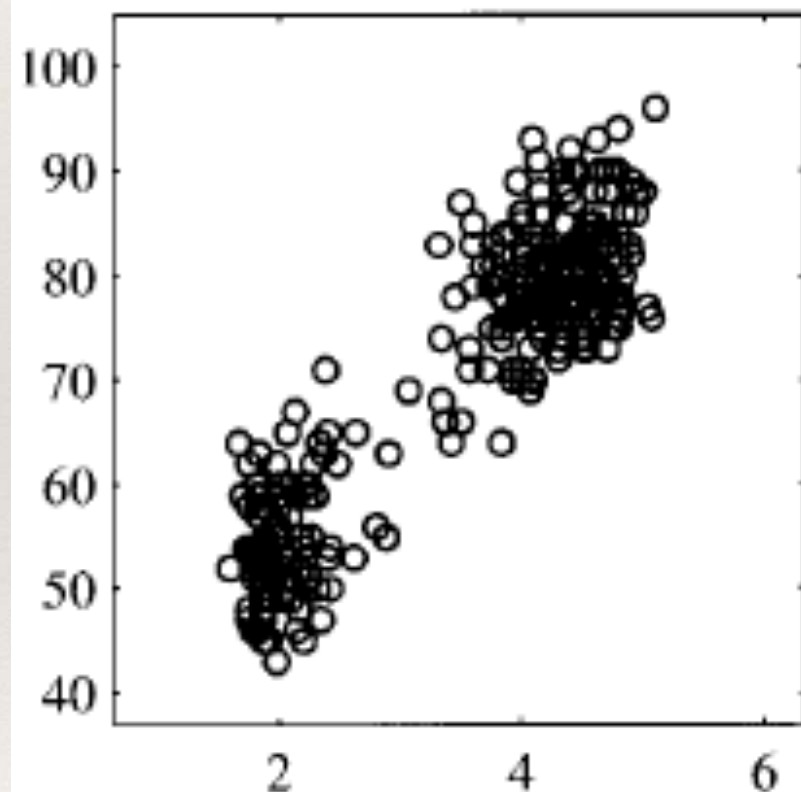


## PCA - Reconstruction

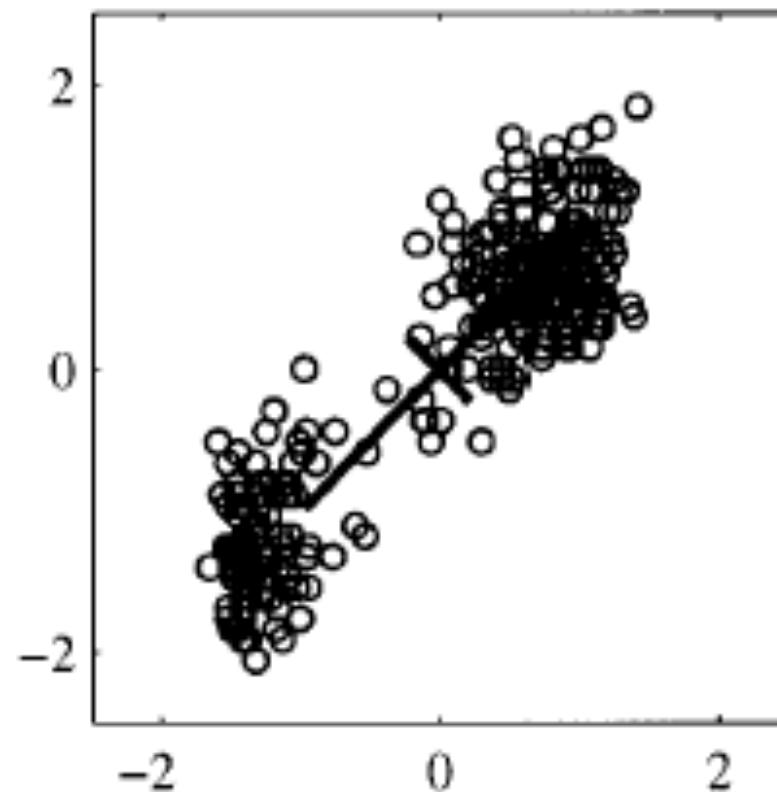


# Whitening the Data

Original Data



Mean Removed data



Whitened data

